

Paige Bryan

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EDUCATION

Purdue University – West Lafayette, IN

May 2027

B.S. in Mechanical Engineering; Minors in Computer Science, Data Science, and Communications

GPA: 3.5/4.0

Honors: Presidential Scholarship, Charles O. McGaughey Leadership Award, Dudley Peak Craig Scholarship, STEM Communication Bridge Scholarship, Mechanical Engineering Outstanding Junior, Gratitude for Service Scholarship

PROFESSIONAL EXPERIENCE

Blue Origin – Renton, WA

Software Engineering Intern

May 2026 – Aug 2026

- Built GitLab CI/CD pipeline to automate LS-DYNA simulations on AWS, using K9 and Slurm to manage compute
- Applied workflow to 4,000+ case Monte Carlo landing analysis for New Glenn and reduced runtime by 66.7%
- Designed React/FastAPI materials database to replace Granta MI, saving \$1.3M+ in annual licensing costs

NASA – Huntsville, AL

Robotics Engineering Intern

Aug 2025 – Dec 2025

- Developed laser-based profiling system with 3D-printed LJ-X8400 mount to map Flat Floor and assess damage
- Redesigned DISCMAN external hardware in Creo to reduce size and achieve operational vacuum for testing on ISS
- Wrote MATLAB program to analyze 1,000+ points of TVAC data and verify health of LASAR robotic subsystems

ExxonMobil – Joliet, IL

Mechanical Engineering Intern

May 2025 – Aug 2025

- Recalculated heat exchanger shell thickness in PV Elite per ASME Section VIII, extending life by 5+ years
- Saved \$350K in replacement costs by exceeding manufacturer estimates through re-evaluation of service conditions
- Performed inspection of piping circuit using OpenSpace LiDAR mapping and IR scans to identify dead legs

John Deere – Dubuque, IA

Advanced R&D Product Engineering Intern

May 2024 – Aug 2024

- Collected 400 million topographical data points using LiDAR and photogrammetry drone scans
- Programmed MATLAB tool to read LAS files and generate occupancy maps for autonomous ADT path planning
- Created app for Advanced R&D Team to streamline project submission and evaluation, targeting 800+ users

LEADERSHIP & VOLUNTEER EXPERIENCE

Purdue Engineering Student Council

Alumni Relations Committee Head

Dec 2024 – Present

- Lead new committee with \$15,000 budget, coordinating 50th-anniversary banquet and tailgate for 100+ alumni

Industrial Roundtable Public Relations Lead

Jan 2024 – Dec 2024

- Organized recruiter-led panel and resume reviews for Prep Night to provide 500+ participants with career advice
- Raised \$35,000 for the Industrial Roundtable Scholarship and collaborated with professors to select recipients

Purdue Mechanical Engineering Ambassadors

Professional Development Chair

Dec 2025 – Present

- Direct 10-person team responsible for planning resume workshops, faculty research panels, and employer site visits
- Engage with over 150 prospective mechanical engineering students each semester at 12+ outreach events

Purdue College of Engineering

Dean's Leadership Scholar

Aug 2024 – Present

- Represent 12,000+ undergraduate engineering students, offering feedback on strategy and new initiatives
- Engage with the Engineering Advisory Council and promote industry partnerships to benefit students

American Society of Mechanical Engineers

Treasurer

Jan 2025 – Aug 2025

- Managed \$30,000 budget across three business accounts, processing all reimbursements and purchase requests
- Conducted comprehensive analysis of past spending to identify trends and optimize future costs

TECHNICAL SKILLS

Software & Robotics: Python, MATLAB, R, C/C++, C#, Java, React/FastAPI, Power Apps, ROS, LiDAR

Design & Simulation: LS-DYNA, LS-OPT, HyperMesh/HyperView, Creo, Siemens NX, Fusion 360, Onshape, GD&T

Cloud & DevOps: AWS, GitLab CI/CD, Slurm, Docker, Linux, Bash, Git/GitHub